RAMBØLL

PROJECT	PROJECT NO.	PREPARED —
—	—	CHECKED —
		APPROVED —

Shear resistance of headed studs

EN 1994-1-1 6.6.3.1

REV / DATE
0.1 · 2026-10-13

DRAFT

PURPOSE
design resistance of a 19 mm headed stud welded through the deck, governed by stud shank failure or concrete crushing, checked against the design shear per stud.

1 Input Parameters

Given quantities as defined in the source document.

Symbol	Value	Unit	Description
d	19	mm	Stud diameter
f_u	450	MPa	Ultimate tensile strength of the stud
h_{sc}	100	mm	
f_{ck}	30	MPa	
E_{cm}	33	GPa	
γ_V	1.25		
P_{Ed}	57.2	kN	

2 Calculation

Derived step by step; each step cites its clause.

Stud height factor

$$\alpha = 1 \text{ if } \frac{h_{sc}}{d} \geq 4, \quad 0.2 \cdot \left(1 + \frac{h_{sc}}{d}\right) \text{ otherwise}$$
$$= 1 \text{ if } \frac{100\ \text{mm}}{19\ \text{mm}} \geq 4, \quad 0.2 \cdot \left(1 + \frac{100\ \text{mm}}{19\ \text{mm}}\right) \text{ otherwise}$$
$$= \mathbf{1}$$

EN 1994-1-1:2004 §6.6.3.1(1), Eqs. (6.20)-(6.21)

Design shear resistance of a headed stud — shank failure

$$P_{Rd,1} = \frac{0.8 \cdot \pi \cdot d^2 \cdot f_u}{4 \cdot \gamma_V}$$
$$= \frac{0.8 \cdot \pi \cdot (19\ \text{mm})^2 \cdot 450\ \text{MPa}}{4 \cdot 1.25}$$
$$= \mathbf{81.66}\ \text{kN}$$

EN 1994-1-1:2004 §6.6.3.1(1), Eq. (6.18)

Design shear resistance of a headed stud — concrete crushing

$$P_{Rd,2} = \frac{0.29 \cdot \alpha \cdot d^2 \cdot \sqrt{E_{cm} \cdot f_{ck}}}{\gamma_V}$$
$$= \frac{0.29 \cdot 1 \cdot (19\ \text{mm})^2 \cdot \sqrt{33\ \text{GPa} \cdot 30\ \text{MPa}}}{1.25}$$
$$= \mathbf{83.33}\ \text{kN}$$

EN 1994-1-1:2004 §6.6.3.1(1), Eq. (6.19)

$$P_{Rd} = \min\left(P_{Rd,1}, P_{Rd,2}\right)$$
$$= \min\left(81.66\ \text{kN}, 83.33\ \text{kN}\right)$$
$$= \mathbf{81.66}\ \text{kN}$$

EN 1994-1-1 6.6.3.1

RAMBØLL

PROJECT
—

PROJECT NO.
—

PREPARED —
CHECKED —
APPROVED —

Shear resistance of headed studs
EN 1994-1-1 6.6.3.1

REV / DATE
0.1 · 2026-10-13

DRAFT

$$h_{eff} = h_{sc} - 0.003\ \text{m}$$
$$= 100\ \text{mm} - 0.003\ \text{m}$$
$$= \mathbf{97}\ \text{mm}$$

EN 1994-1-1 6.6.3.1

$$UR = \frac{P_{Ed}}{P_{Rd}}$$
$$= \frac{57.2\ \text{kN}}{81.66\ \text{kN}}$$
$$= \mathbf{0.7005}$$

EN 1994-1-1 6.6.3.1

3 Verification

Acceptance criteria for the governing results.

OK	$UR = 0.7005 \leq 1$